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**BIT 2324 GEOGRAPHICAL INFORMATION
SYSTEMS**

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This presentation is intended to be covered within one week. The notes, examples and exercises should be supplemented with a good textbook. Most of the exercises have solutions/answers appearing elsewhere and accessible by clicking the green **Exercise** tag. To move back to the same page click the same tag appearing at the end of the solution/answer.

Errors and omissions in these notes are entirely the responsibility of the author who should only be contacted through the **Department of Curricula & Delivery (SODeL)** and suggested corrections may be e-mailed to elarning@jkuat.ac.ke



LESSON 7

Cartographic Design

Learning outcomes

By the end of this topic you should be able to;

- *Identify security threats to databases*
- *Discuss multi-level database security*



7.1. Introduction

7.2. Principles of cartographic design

1. Use of color: The primary function of color is to make information on a map visually distinguishable

Dimensions of color

- Hue: dominant wavelength (i.e., color) - Hue is generally used to indicate qualitative (nominal) differences across the map area
- Value: how light or dark a given hue is
- Saturation: purity of hue (range of reflected wavelengths - Value and saturation are typically used to represent quantitative (ordinal, interval, or ratio) differences across the map area



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2. Use of text: Descriptive text is used to give a map its title, to explain the legends and label features. Text is stored as either symbols of a coverage or annotation coverage

Text characteristics:

- Family Arial vs Helvetica
 - Style (face) bold vs italic
 - Font 32 point vs 48 point
 - Color black vs blue
3. Symbols and symbol sets: A symbol is a graphic pattern that is used to represent a feature on a map.
 - Marker symbol: points and nodes (.mrk)
 - Line symbols: arcs (.lin)
 - Shade symbols: polygons (.shd)
 - Text: annotation (.txt)



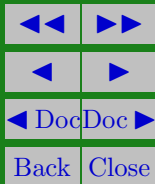
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In symbol sets, each symbol must be programmed individually, so GIS software packages usually supply the user with predefined symbol sets that can be edited and customized.

4. Map-to-page transformation

- Physical page: actual surface of display medium
- Graphics page: portion of physical page where map is drawn
- Map limits: portion of graphics page where coverage features are drawn
- Map extent: area of the earth's surface to be displayed (in actual ground units)

Map composition





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- Map layout design: cartography is both a science and an art, therefore subjectivity and creativity play an important role in the aesthetic quality of a map
- Geographical contents: elements to be included are governed by:
 - Map theme i.e., land use
 - Map coverage i.e. Illinois vs. U.S.
 - Map scale i.e. level of generalization
- Label placement: Good label placement ensures readability and enables the reader to associate labels with the map elements that they describe.

7.2.1. Geographic contents to be considered:

- Visual balance: location/proportionality



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- Visual clarity: generalization
- Visual hierarchy: font & symbol size
- Visual contrast: use of color and shading
- Context: selection of data layers
- Text: annotation of features

7.2.2. Cartographic generalization

1. Why? When geographic data are gathered at a scale that is larger than the scale at which they are presented, it is necessary to reduce the complexity of the data to make the resulting map more aesthetically pleasing. Generalization ensures that geographic data are presented at a scale appropriate to the purpose of the map and the application



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requirements of the user.

2. When?

- Simplification
- Smoothing
- Aggregation
- Amalgamation
- Merging
- Collapse
- Refinement
- Typification
- Exaggeration
- Displacement
- Classification

3. How?



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- **Congestion:** too many features in too little space
- **Coalescence:** features touch due to inadequate symbolization
- **Conflict:** feature symbol incompatible with background
- **Complication:** data from different sources or at different scales or levels of tolerance
- **Inconsistency:** Generalization applied in a nonuniform fashion across map
- **Imperceptibility:** Feature falls below minimum resolution of map

EXERCISE 1. 🖱️ Explain the advantages of role base access control in multi level databases.





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Example . Giving suitable examples, differentiate between credentials and population as used in multilevel databases.

Solution: Apulation or the average income in the population.

□

EXERCISE 2.  Explain covert channel as used in multilevel data bases.



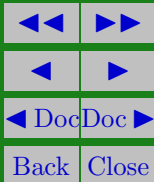
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Solutions to Exercises

Exercise 1. RBAC ensures that only authorized users arations.

Exercise 1





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Exercise 2. A covert chents or processes.

Exercise 2

